

Prerequisites on Windows Server and Client before Implementing Privilege Escalation Use Case

Summary

In this document, there are steps to verify and enable required audit policies and then test by simulating an Privilege Escalation attack by creating a “testuser” and then clean the previous configs and creating a domain user for later use in the use case.

1. Ensure the following Event IDs are visible in Event Logs

Event ID	Meaning
4732	User added to local group
4728	User added to global group
4672	Admin privileges assigned
4624	Successful login
4625	Failed login

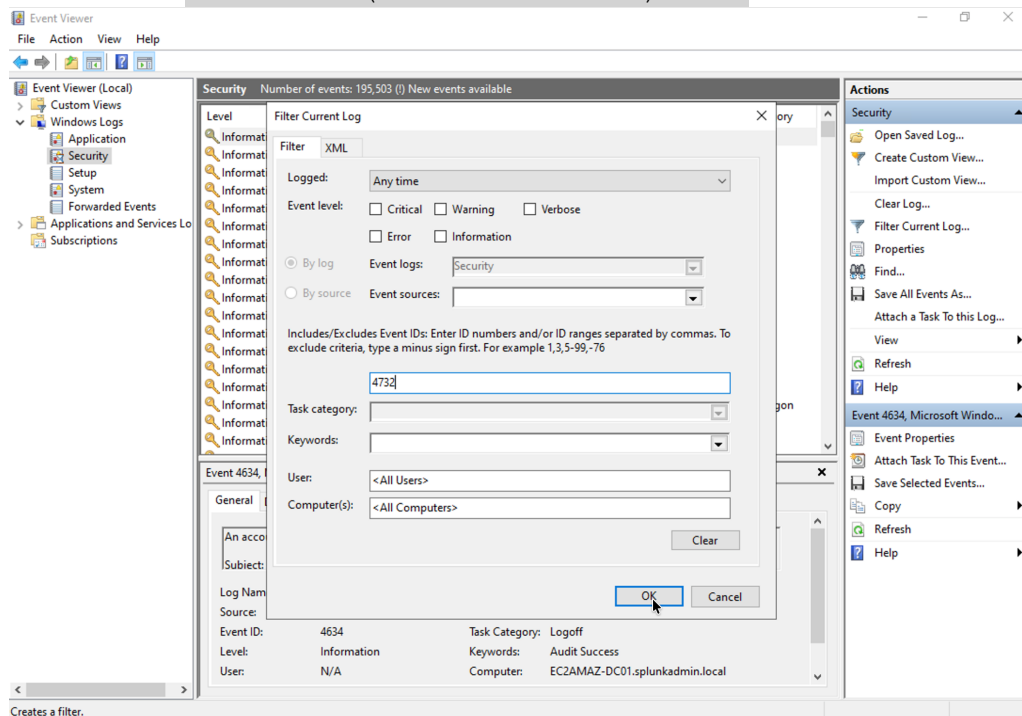
1) In Windows Server,

Go to `Start > Event Viewer > Windows Logs > Security`

There when the event logs appear,

Click `Filter Current Log...`

Enter the Event IDs (mentioned above) > OK

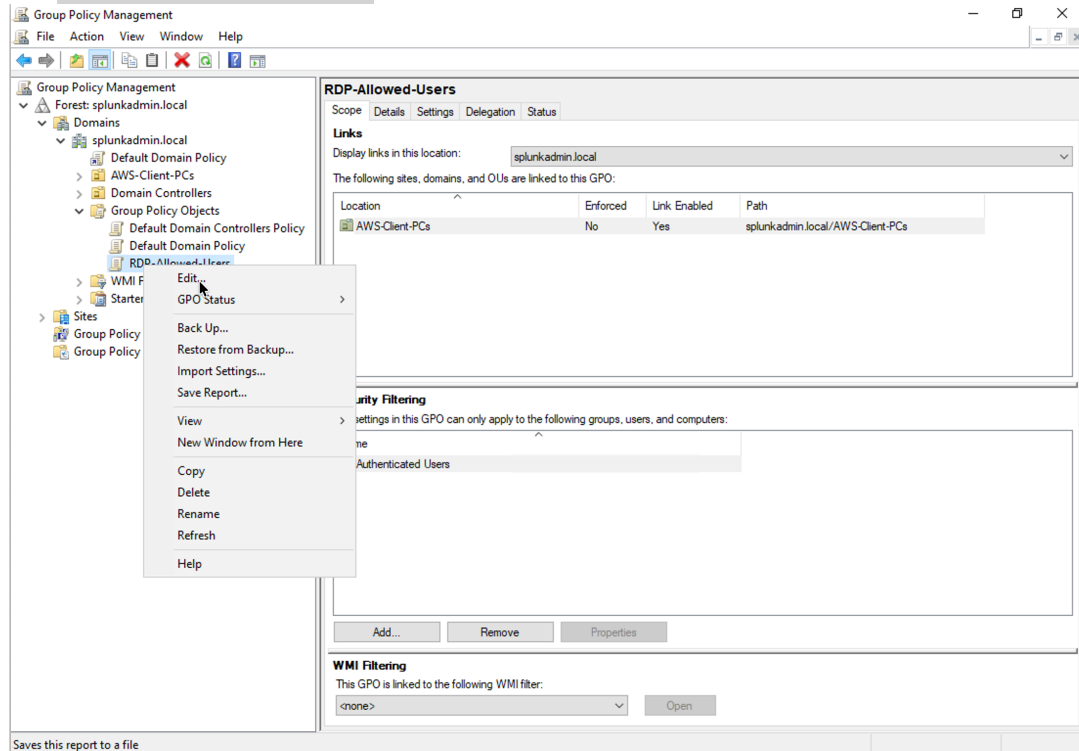


Note: If **no logs** appear related to Event IDs, then **audit policies must be enabled** for logging of those Event IDs when the events occur; move to **Step 2**. If all Event IDs are **visible** move to **Step 3**.

2. Enable the Audit Policies

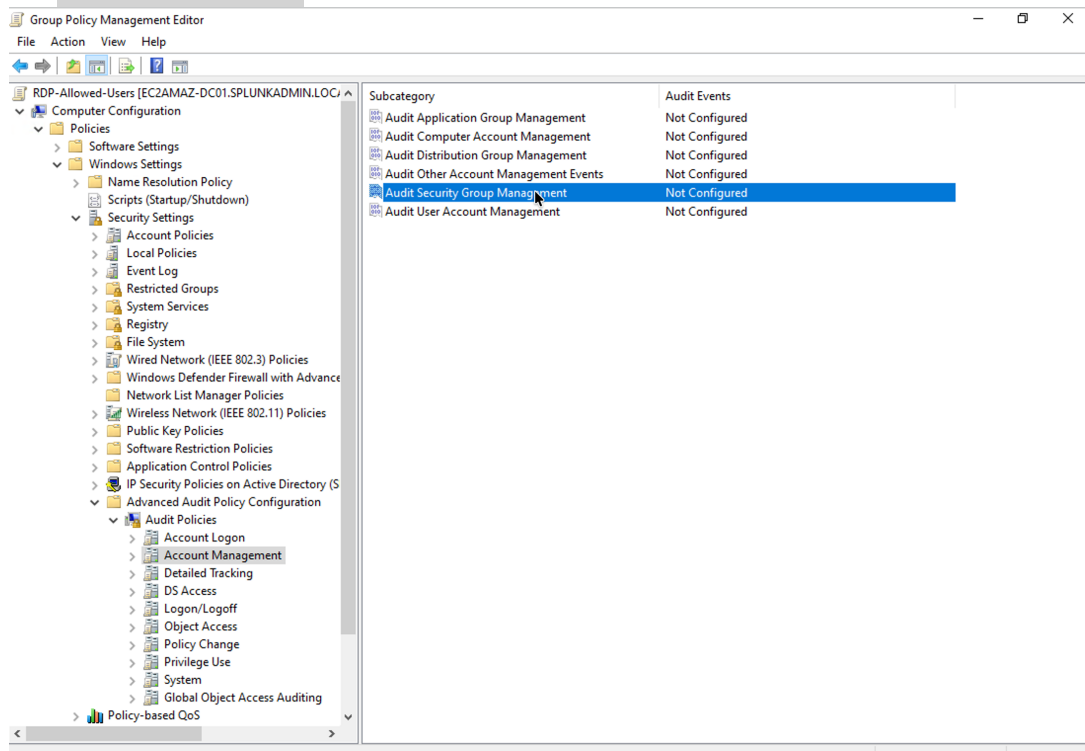
1) Open Group Policy Management (gpmc.msc)

Navigate to Forest:<yourforest.local> > Domains > <yourdomain.local> > Group Policy Objects > right-click <Your-GPO> > Edit...



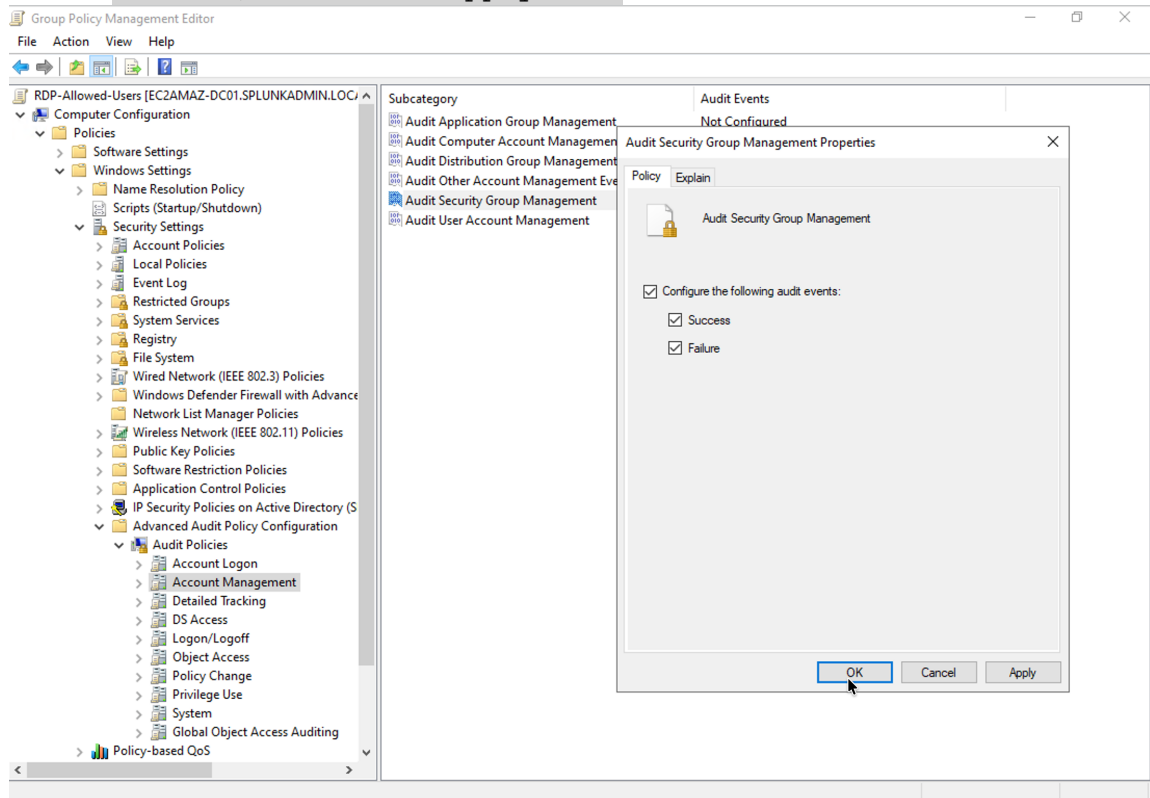
2) Go to:

Computer Configuration > Policies > Windows Settings > Security Settings > Advanced Audit Policy Configuration > Audit Policies

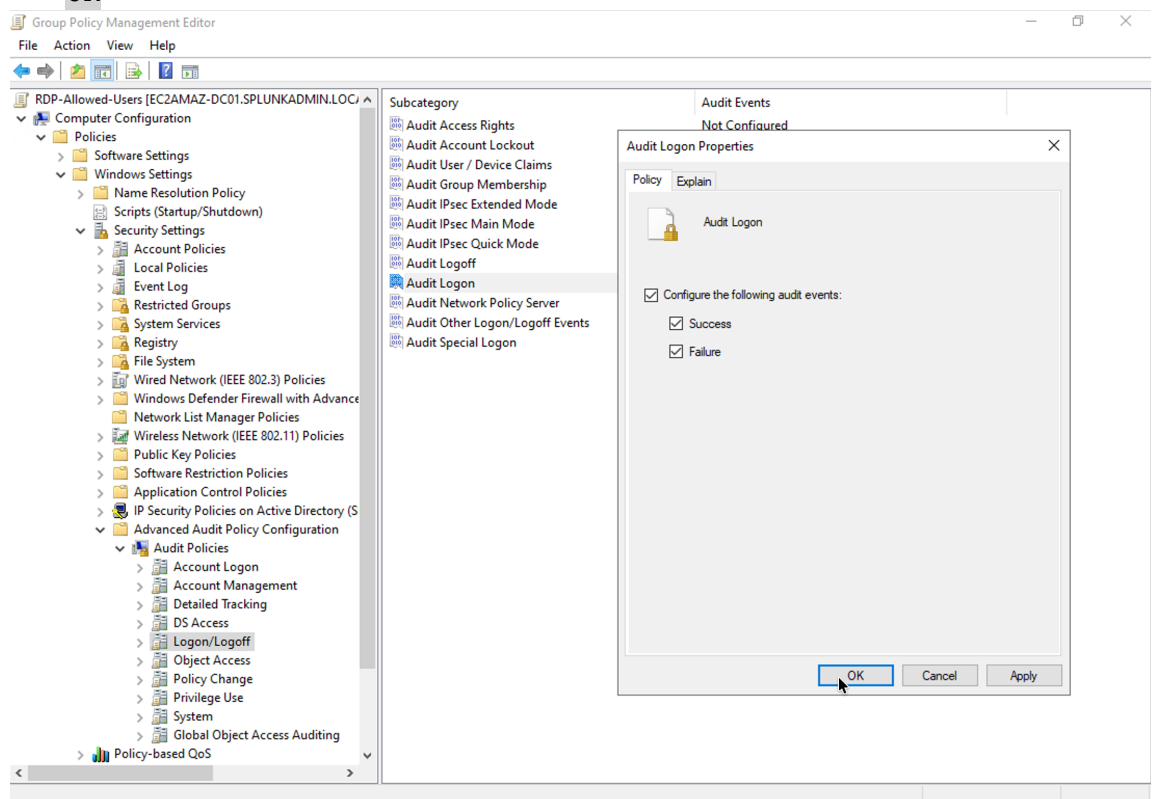


3) Enable the following:

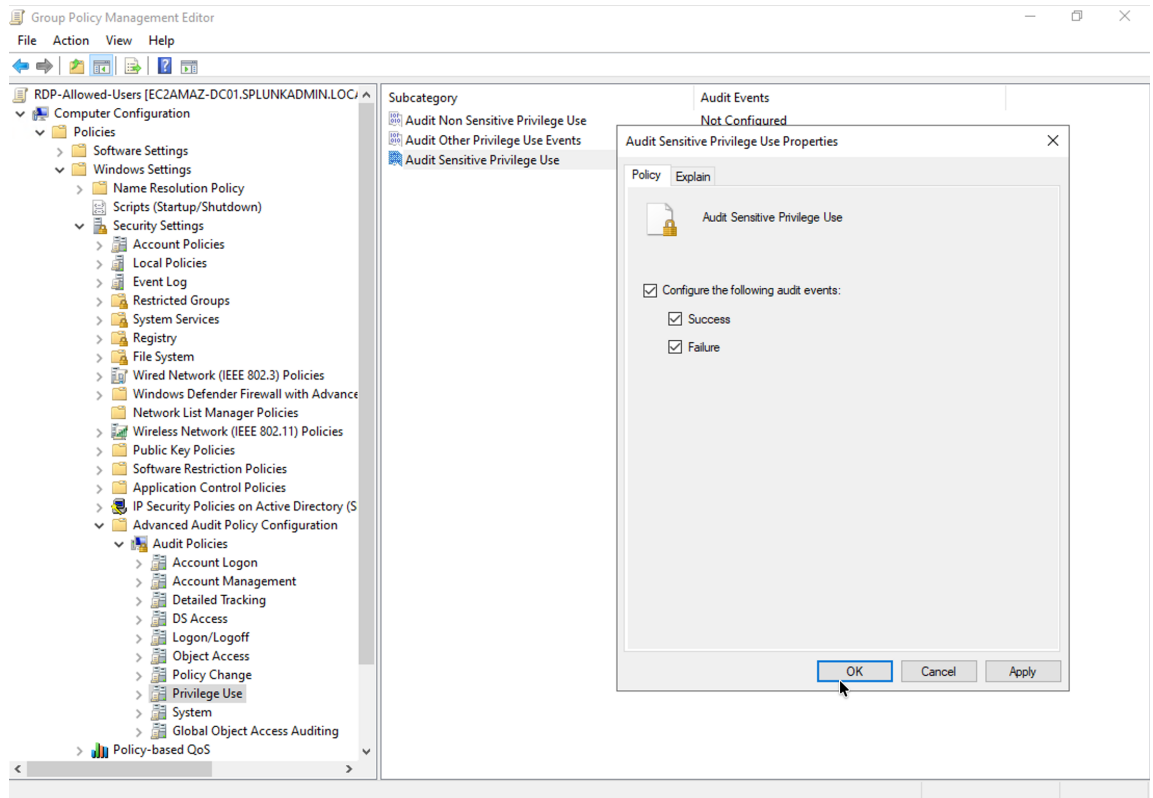
- i) Account Management > Audit Security Group Management: Success, Failure > Apply > OK



- ii) Logon/Logoff > Audit Logon: Success, Failure > Apply > OK



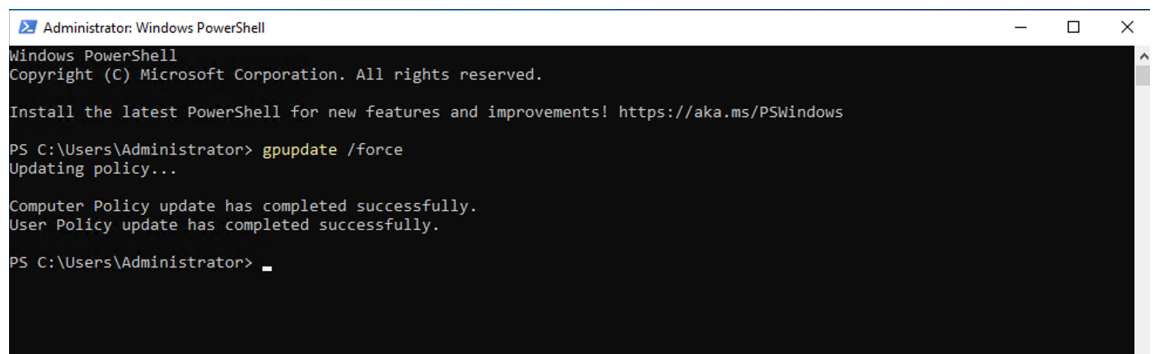
- iii) Privilege Use > Audit Sensitive Privilege Use: Success, Failure > Apply > OK



4) Update the Group Policy

- i) In **Windows Server**,
Open PowerShell and Run:

```
gpupdate /force
```



- ii) In **Windows Client**,
Update the Group Policy via PowerShell:

```
gpupdate /force
```

5) Verify on Event Viewer

- i) Simulate a **Privilege Escalation** by adding a **testuser** to local group **Administrators**
In **Windows Server**,
Open PowerShell & Run:

```
net user testuser <password> /add
net localgroup Administrators testuser /add
```

```
Administrator: Windows PowerShell
Windows PowerShell
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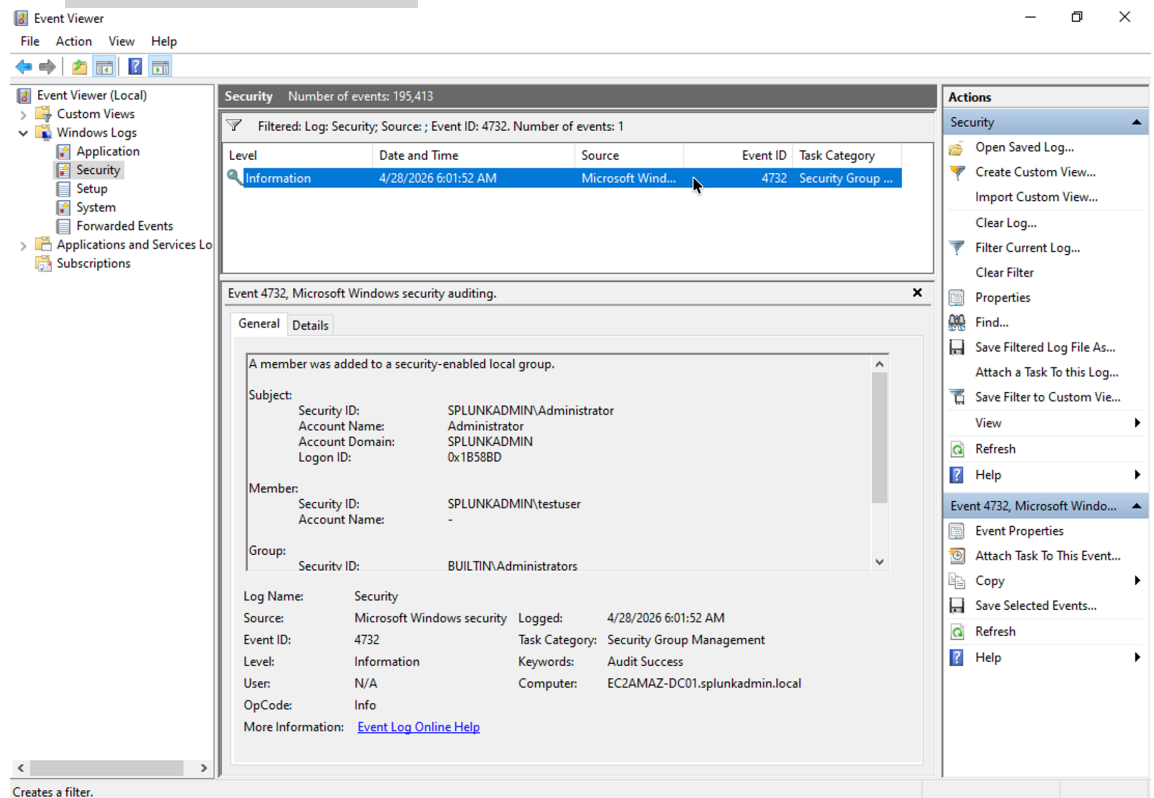
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> net user testuser Password123! /add
The command completed successfully.

PS C:\Users\Administrator> net localgroup administrators testuser /add
The command completed successfully.

PS C:\Users\Administrator> _
```

- ii) Open **Event Viewer** and test
Click **Filter Current Log...** under **Windows Logs > Security**
Event IDs: 4732 > OK



3. Delete the testuser

Open **PowerShell** & Run:

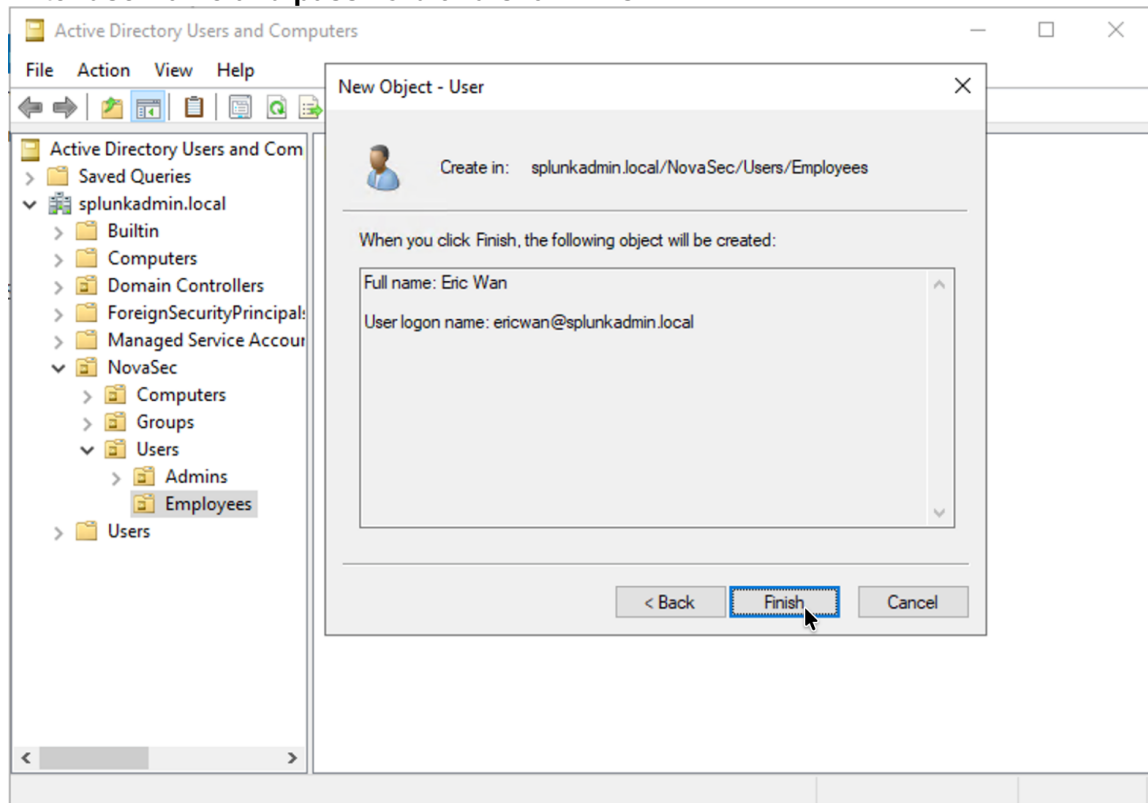
```
net localgroup Administrators testuser /delete  
net user testuser /delete
```

4. Create a Domain User on Server

Go to **NovaSec > Users > Employees**

Right-click > New > User

Enter **username and password** and Click **Finish**



Next Steps

Simulating a Privilege Escalation attack on Windows Server and Client for Log Validation – Refer **23_Simulating_PrivEsc_Attack_WinServer_WinClient_V1.pdf**.